4.1 Purpose of Macros

Macros are defined using "#define" and are expanded by the preprocessor before compilation. They are useful for repeated calculations such as converting rupees to paise and finding the larger of two values.

4.2 Macro Definitions

The "SQUARE(x)" macro calculates the square of a value. Parentheses are necessary to avoid unexpected results when expressions are passed as arguments.

The "MAX(a, b)" macro returns the larger of two values. However, an argument may be evaluated more than once, which can cause errors when using expressions such as "i++".

4.3 Macro Pitfalls

Without proper parentheses, "SQUARE(a + 1)" may produce an incorrect result because of operator precedence. Writing "#define SQUARE(x) ((x) * (x))" fixes this problem.

Macros such as "MAX(a, b)" can evaluate an argument twice. A function is safer when arguments have side effects.

4.4 Preprocessor and Complexity Analysis

The command "gcc -E program.c" displays the preprocessed source code after macro expansion.

Macros are expanded wherever they are used, which can increase code size when used many times. A function may require a function call, but a static inline function can allow the compiler to optimize the call. Both approaches generally perform constant-time calculations for simple expressions.

```c
#include <stdio.h>

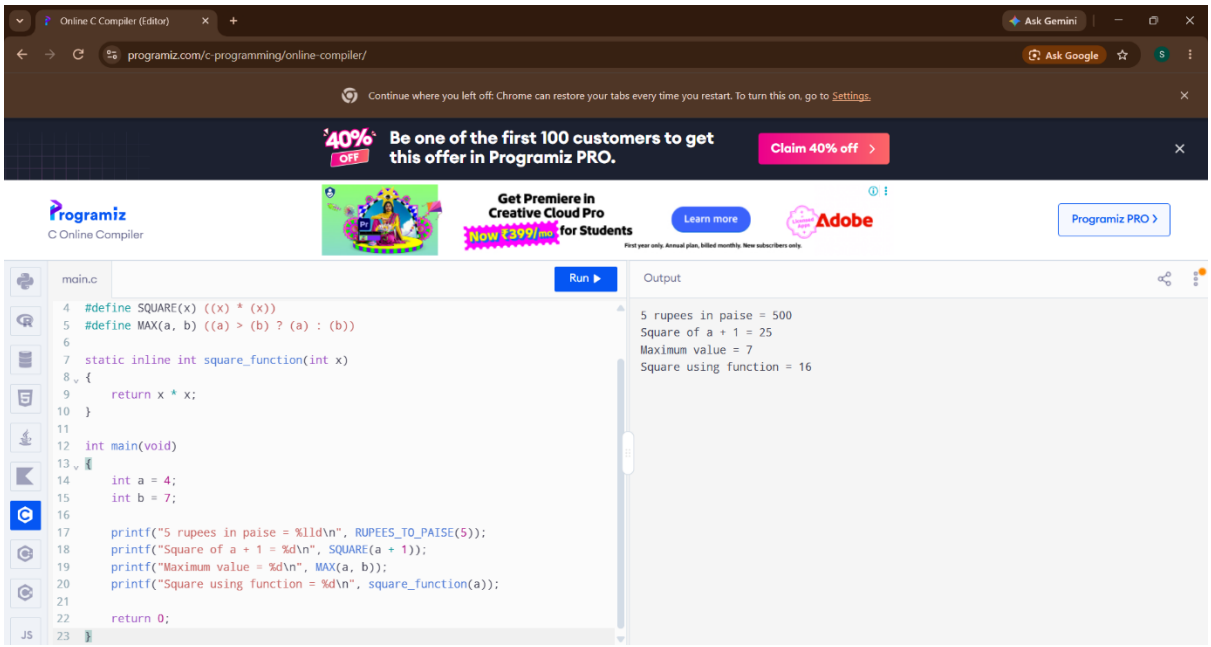
#define RUPEES_TO_PAISE(r) ((long long)(r) * 100)
#define SQUARE(x) ((x) * (x))
#define MAX(a, b) ((a) > (b) ? (a) : (b))

static inline int square_function(int x)
{
    return x * x;
}

int main(void)
{
    int a = 4;
    int b = 7;

    printf("5 rupees in paise = %lld\n", RUPEES_TO_PAISE(5));
    printf("Square of a + 1 = %d\n", SQUARE(a + 1));
    printf("Maximum value = %d\n", MAX(a, b));
    printf("Square using function = %d\n", square_function(a));

    return 0;
}
```


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main.c
Run
Output
#define SQUARE(x) ((x) * (x))
#define MAX(a, b) ((a) > (b) ? (a) : (b))
static inline int square_function(int x)
{
return x * x;
}
int main(void)
{
int a = 4;
int b = 7;
printf("5 rupees in paise = %lld\n", RUPEES_TO_PAISE(5));
printf("Square of a + 1 = %d\n", SQUARE(a + 1));
printf("Maximum value = %d\n", MAX(a, b));
printf("Square using function = %d\n", square_function(a));
return 0;
}
5 rupees in paise = 500
Square of a + 1 = 25
Maximum value = 7
Square using function = 16